

Rancho Zhao

Senior Backend Engineer · Go / Python

✉ 1006959160@qq.com

☎ 18883855775

🌐 blog.ranchocooper.com

📁 github.com/RanchoCooper



Summary

Senior backend engineer with **9 years** of experience at leading companies including **Alibaba Local Services (Ele.me)**, **Binance**, and **Liulishuo (Laix)**. Deep expertise in Go / Python tech stacks, with hands-on experience in high-concurrency service architecture and Domain-Driven Design. Over the past two years, focused on **AI Agent development** and **quantitative trading systems**, bridging cutting-edge technology with business value. Proven cross-team collaboration and technical leadership—led engineering teams to deliver mission-critical projects.

Technical Skills

Go

Python

Java

Gin

GORM

Flask

MySQL

PostgreSQL

Redis

Docker

Linux

Git

Domain-Driven Design

Hexagonal Architecture

TDD

AI Agent Development

LLM Applications

Quantitative Trading

Education

Chongqing University of Posts and Telecommunications · B.E. in

2013.09 — 2017.06

Computer Science and Technology

Work Experience



Wenzhou Huanteng Technology · Senior Backend Engineer

2026.03 — 2026.06

- Led backend development for the Feiying AI platform, powering AI image generation, AI video generation, and Agent conversational features across multiple business lines
- Integrated Alibaba Wanxiang, Doubao, Wengao Cloud and other model providers; built a unified model invocation layer with async task scheduling



Avenir Group (Shenzhen) · Senior Backend Engineer

2024.11 — 2025.04

- Architected the backend system for the investment research platform, supporting **millions of daily** financial data collection and processing tasks
- Built a complete data pipeline (Airflow + Doris + Django) covering collection, storage, and visualization, significantly improving research data throughput
- Developed Grafana dashboards for real-time monitoring and visualization of key business metrics
- Supported the group risk-control department by designing and implementing an IP proxy pool solution to ensure data collection stability and compliance



Binance · Senior Backend Engineer

2021.09 — 2023.06

- Core engineer for Themis, an A/B experimentation platform serving **tens of millions of users** for traffic splitting and experiment analysis
- Designed and implemented a three-tier caching architecture for the experiment routing engine, achieving millisecond-level matching latency
- Built a visual experiment configuration platform, reducing the operational cost of experiment iteration
- Founded the company-wide Go Hexagonal internal framework (DDD + Hexagonal Architecture), adopted across multiple business lines; integrated CLI scaffolding to accelerate new project bootstrapping
- Led a two-engineer team as tech lead, responsible for task assignment, code review, and mentorship



Liulishuo (Laix) · Senior Backend Engineer

2019.11 — 2021.01

- Responsible for backend architecture design and core feature development for multiple education products, including small-group classes and pronunciation training
- Led database sharding and partitioning for tables with hundreds of millions of rows, achieving significant read/write performance gains
- Built a unified OSS authentication service from scratch, ensuring consistent and secure user access across product lines
- Introduced Prometheus + Grafana observability stack, enabling full-stack service monitoring
- Developed Go-based Hive data ingestion jobs for scheduled structured data warehousing and data pipeline governance



Alibaba Local Services (Ele.me) · Senior Backend Engineer

2017.07 — 2019.11

- Developed and iterated on core logistics systems including pick-up/delivery separation, last-mile delivery, and autonomous delivery
- Led architecture upgrade and SOA migration (Spring + Pizza Gateway) of the last-mile delivery system, supporting multi-region active-active deployment
- Designed integration standards for autonomous delivery; led the integration of autonomous vehicles and delivery robots into the existing last-mile system, coordinating with hardware vendors on elevator adaptation
- Migrated legacy Python systems to Java using DDD, significantly improving maintainability and extensibility
- Implemented an intelligent route planning algorithm with Flask, improving delivery efficiency by 25%; pilot testing in targeted districts yielded 30% manpower efficiency gain and 20% cost reduction per order
- Drove unit test coverage improvement across the logistics department and served as a mentor for new hires



Zhihu · Python Engineer (Intern)

2017.01 — 2017.04

- Contributed to the development and maintenance of the community digest module; participated in technical design reviews

| Personal Projects & Open Source



Mood Map Pro — Anonymous Social Mini Program

WeChat Mini Program with **Go (Gin)** backend, PostgreSQL + WebSocket real-time communication. Features: mood posting with geo-heatmap, smart matching by mood type and city, time-limited 5-minute chat sessions with extension, crisis keyword detection with automatic mental health hotline referral.



WealthWise — Personal Finance Mini Program

Native WeChat Mini Program with client-side architecture (local Storage + Canvas charts). Features: primary/side income tracking, expense management, multi-goal financial planning, wealth trend charts, transaction history with undo, customizable theme poster generation and sharing.



Go Hexagonal + Py Hexagonal [GitHub ↗](#)

Go / Python project templates based on Hexagonal Architecture and Domain-Driven Design, adopted across multiple business lines at Binance. Achieves clear separation of business logic and infrastructure through dependency inversion and interface segregation, improving testability and maintainability.



Quants / Py-Quantix — Quantitative Trading System [GitHub ↗](#)

Personal quantitative trading system covering data collection, strategy backtesting, signal generation, and live trading. Under active development.



Hexo Blog Theme [GitHub ↗](#)

Custom Hexo theme powering blog.ranchocooper.com.



Open Source Contributions

[structs](#) — improved tag functionality; [Vibora](#) — fixed routing bugs and improved documentation.